Learning in Multiple Spaces: Prototypical Few-Shot Learning With Metric Fusion for Next-Generation Network Security

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***Abstract*—As next-generation communication networks increasingly rely on AI-driven automation, ensuring robust and secure intrusion detection becomes critical, especially under limited labeled data. In this context, we introduce *Multi-Space Prototypical Learning* (MSPL), a few-shot intrusion detection framework that improves prototype-based classification by fusing complementary metric-induced spaces (Euclidean, Cosine, Chebyshev, and Wasserstein) via a constrained weighting mechanism. MSPL further enhances stability through Polyak-averaged prototype generation and balanced episodic training to mitigate class imbalance across diverse attack categories. In a few-shot setting with as few as 200 training samples, MSPL consistently outperforms single-metric baselines across three benchmarks: on CICEVSE Network2024, AUPRC improves from 0.3719 to 0.7324 and F1 increases from 0.4194 to 0.8502; on CICIDS2017, AUPRC improves from 0.4319 to 0.4799; and on CICIoV2024, AUPRC improves from 0.5881 to 0.6144. These results demonstrate that multi-space metric fusion yields more discriminative and robust representations for detecting rare and emerging attacks in intelligent network environments.**

***Index Terms*—Few-shot learning, network intrusion detection, metric-based learning, multi-space prototypical learning.**

I. Introduction

THE integration of artificial intelligence (AI) into next-generation communication networks has revolutionized threat detection. However, it has also introduced new vulnerabilities that adversaries increasingly exploit. The rapid evolution of cyber threats poses a critical challenge in cybersecurity. Attackers now leverage sophisticated techniques to compromise AI-driven systems and exploit model weaknesses [1]. While traditional network intrusion detection systems (NIDS) remain effective against known attack signatures, they struggle to detect emerging, rare, or zero-day attacks. This challenge is especially acute when labeled data is scarce and attack behaviors evolve dynamically [2], [3], [4]. This growing gap highlights the urgent need for adaptive, data-efficient, and resilient detection frameworks. Such frameworks must operate reliably in complex, AI-augmented network environments.

Few-shot learning (FSL) has emerged as a promising solution to these challenges by enabling models to generalize from minimal labeled data using meta-learning principles [5], [6], [7]. In the context of federated and distributed AI systems, FSL offers a pathway to privacy-preserving and scalable intrusion detection. However, early applications of FSL in cybersecurity have revealed persistent issues—including data imbalance, limited scalability, and poor generalization to real-world network conditions [8], [9]. These limitations are further exacerbated by the rise of generative AI, model inversion, and poisoning attacks, which threaten both data integrity and model trustworthiness. Addressing these concerns requires a new generation of secure and trustable AI-based systems that can withstand adversarial manipulation while maintaining high detection performance in dynamic network environments.

Recent advancements have sought to overcome these challenges through innovative frameworks and methodologies. For instance, Ma et al. [10] introduced a few-shot IoT attack detection framework that integrates adaptive loss weighting to enhance detection performance. Similarly, hybrid and metric-based approaches, such as those proposed by Liang et al. [11] and Zhou et al. [12], have shown promise in detecting anomalies in complex environments like industrial IoT and cyber-physical systems. Despite these advancements, critical gaps still need to be addressed, including the inability to robustly handle diverse attack scenarios and limited scalability for real-world deployment.

We present a multi-space prototypical learning (MSPL) framework for few-shot attack detection that leverages complementary metric spaces - Euclidean, Cosine, Chebyshev, and Wasserstein distances - to capture diverse attack pattern properties [13], [14]. Our approach extends beyond Martinez et al.'s dual-space methodology [15] to capture attack pattern characteristic through distinct geometric and statistical perspectives. In addition to that, our framework enables the use of Polyak-averaged prototype generation and balanced episodic training to improve detection stability and representation across attack types.

The main contributions of this work include:

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- A novel multi-space integration that combines four complementary distances through a constrained weighting scheme, demonstrating consistent improvements over single-metric baselines.
- Comprehensive validation across three benchmark datasets (CICEVSE2024 [16], CICIDS2017 [17], and CICIoV2024 [18]) demonstrating the framework's generalizability and superior performance in detecting both high-profile and low-profile attacks, particularly excelling at identifying rare and zero-day threats with as few as 200 training samples.
- A balanced episodic training strategy that ensures equal representation of attack classes through stratified sampling and controlled repetition, specifically addressing the challenge of imbalanced cybersecurity datasets.
- Polyak-averaged prototype generation that maintains exponential moving averages of prototypes across training episodes, substantially reducing representation variability and improving convergence.

Collectively, these contributions form a cohesive MSPL framework that is both practically deployable and theoretically grounded for next-generation network settings. In turn, tackling these challenges not only progresses the forefront of few-shot intrusion detection but also lays the groundwork for creating more resilient and adaptable cybersecurity systems. The integration of complementary learning paradigms highlights the potential for future research to explore hybrid methodologies that extend the framework's capabilities to multimodal datasets and real-time detection systems [10], [19].

The remainder of this paper is organized as follows: Section II reviews related work; Section III introduces the foundations of multi-space few-shot attack detection; Section IV presents MSPL; Section V analyzes its theoretical properties; Section VI reports the experimental results; and Section VII concludes the paper.

II. Related Work

FSL has emerged as a powerful approach to address data scarcity in network intrusion detection [5], [6], [20]. Early work by Chowdhury et al. [21] demonstrated the effectiveness of deep learning for detecting common attacks, though scalability remained a challenge. Iliyasu et al. [22] improved discriminative power using supervised autoencoders, particularly for low-profile and imbalanced attacks. Other studies applied FSL with prototypical networks [8], FS-IDS for imbalanced data [9], and IoT attack detection via self-supervised learning and adaptive loss weighting [10]. While these methods enhance generalization, they rely on a single similarity space, limiting expressiveness under distributional shifts. In contrast, our MSPL framework fuses four complementary metrics through constrained weighting, enabling richer prototype comparisons under data scarcity.

Generative and retrieval-based strategies have also been explored. Aharon et al. proposed a GAN-inspired framework for API attack detection [23] and a classification-by-retrieval approach leveraging embeddings [24], both addressing dynamic API threats. Compared to these, MSPL strengthens prototype geometry by integrating multiple metric perspectives. Further advancements include FSL for SCADA [25], industrial IoT [11], and cyber-physical systems [12], as well as meta-learning [26], class-incremental learning [27], space-decoupled prototypes [28], and graph-based methods [29], [30].

Unlike approaches that modify the embedding architecture (e.g., space decoupling or graph structure), MSPL improves robustness by enriching the distance computation itself via multi-metric fusion.

A. Metric-Based Learning in Cybersecurity

The metric space is critical in FSL for intrusion detection. While traditional approaches rely on Euclidean or cosine metrics, recent studies explore advanced spaces to improve performance. Xu et al. [31] apply meta-learning with task-specific adaptations, and Tian et al. [13] use the Wasserstein metric for spoofing detection in WiFi systems. These works show that different metrics capture distinct notions of similarity

Unlike such single-metric formulations, MSPL integrates Euclidean, Cosine, Chebyshev, and Wasserstein distances concurrently, allowing the decision rule to benefit from both pointwise and distributional discrepancies.

In [12], a Siamese network-based approach integrates multiple metrics for anomaly detection in industrial environments. While Siamese-style designs can combine similarity cues, they are not designed as a constrained multi-space prototype fusion mechanism for few-shot intrusion detection. MSPL complements this line of work by formalizing multi-metric fusion around prototypical representations and stabilizing learning across episodes.

Additionally, Miao et al. [32] proposed a Siamese prototypical network (SPN) incorporating out-of-distribution detection for traffic classification. This method is particularly adept at identifying anomalous traffic patterns that deviate significantly from known attack profiles, demonstrating the utility of integrating prototypical networks with robust metric-based evaluation mechanisms. Autoencoder-based approaches have also been explored for metric-driven FSL. He et al. [33] utilized deep autoencoders to capture rich feature representations for malicious traffic detection. Their approach combines feature learning with metric-based classification, achieving high accuracy in few-shot scenarios. Further, Vijayakanthi et al. [14] introduced a differential metric-based methodology for non-profiled side-channel analysis, primarily in cryptographic contexts. While their work is focused on a different domain, it provides valuable insights into designing adaptive metric-learning systems that could be applied to intrusion detection.

B. Hybrid and Multi-Space Approaches

Hybrid and multi-space learning approaches have shown promise in addressing the limitations of single-metric models. For instance, ProtÉdge [19] combines multiple metrics to improve detection accuracy in software-defined networks, and [34] applies cross-network meta-learning for unseen malware variants. The effectiveness of metric fusion is further demonstrated in [15], where a dual-space prototypical network enhances DDoS attack detection by integrating complementary metric spaces. Compared to prior fusion efforts that typically combine a small number of metrics/spaces (e.g., dual-space designs), MSPL broadens the fusion to four complementary metrics and couples it with a constrained weighting scheme to maintain stable prototype-based decision boundaries.

Further studies, such as [22] and [35], focus on combining supervised and unsupervised learning techniques to improve generalization. Additionally, [36] proposes a model-agnostic framework that integrates generative models with metric-based learning, significantly improving the detection of low-profile attacks. Graph neural networks (GNNs) have also been utilized for few-shot anomaly detection [37], demonstrating the potential of graph-based metrics in capturing structural patterns. In contrast to hybrid pipelines that expand training signals via auxiliary objectives (e.g., generative or contrastive learning), MSPL targets the metric layer directly, using multi-metric fusion and episode-level stabilization to improve few-shot robustness.

Qin et al. [38] developed a meta-learning-based framework for zero- and few-shot face anti-spoofing. Although their focus is on biometric security, the proposed meta-model offers insights into how meta-learning can generalize across unseen attack scenarios, a principle that could be extended to intrusion detection systems. More efforts include FewM-HGCL [39], which employs contrastive learning on heterogeneous graphs for malware detection, and utilizes graph contrastive learning for classifying network flow attacks. These approaches underscore the importance of leveraging diverse feature spaces and learning paradigms to tackle the evolving landscape of cyber threats.

C. Uniqueness of This Paper

This paper introduces the MSPL framework, which combines four complementary metric spaces, Euclidean, Cosine, Chebyshev, and Wasserstein, via a constrained fusion of normalized distances (z-score with clipping) with simplex weights ($w_m \geq 0$, $\sum_m w_m = 1$), preventing scale dominance across metrics and yielding a well-behaved combined distance. Unlike existing single-metric or dual-metric prototypical baselines, MSPL uses this metric-preserving fusion to exploit distinct topological cues under data scarcity. MSPL further incorporates two stabilizers tailored to few-shot NIDS: (i) Polyak/EMA-based prototype generation to reduce episode-to-episode embedding variance, and (ii) a balanced episodic sampling strategy that enforces per-episode class parity (with controlled repetition when needed). Together, these design choices lead to more robust detection of both high-profile and low-profile attacks, with consistent gains on rare/zero-day categories.

III. Foundations of Multi-Space Learning for Few-Shot Attack Detection

Conventional metric-based FSL frameworks typically employ Euclidean distance to quantify relationships between samples. While this is effective in many applications, this approach imposes a fixed topological assumption on the data, potentially overlooking critical variations in attack patterns. In contrast, cyber threats manifest in diverse ways, ranging from subtle deviations in statistical distributions to structural anomalies in network traffic, requiring a multi-faceted approach.

Single metric-based methods, like regular prototypical networks, often fail to capture the full complexity of attack patterns due to:

- **Attack variability across feature representations:** Certain attacks (e.g., adversarial traffic) exhibit directional similarity in feature space, while others (e.g., protocol abuse) differ in absolute magnitude.
- **Class distribution skewness:** Network intrusion datasets are inherently imbalanced, with some attack types significantly underrepresented. A single distance metric can overfit to dominant classes while failing to generalize to rare attack instances.
- **Data geometry and structure:** Some attack types are better differentiated based on distributional discrepancies (e.g., statistical shifts), while others require pointwise comparisons based on individual feature deviations. In fact, critical discriminative features can alternate between localized anomalies (single-feature spikes detectable through Chebyshev distance) and systemic distribution shifts (Wasserstein-visible pattern deformations).

By introducing multiple metric spaces, we can mitigate the biases inherent in monolithic learning and enhance the ability to differentiate between diverse and overlapping traffic patterns across multiple threat categories. Specifically, integrating Euclidean, Cosine, Chebyshev, and Wasserstein distances enables the framework to capture local and global structural variations in attack data.

A. Complementary Roles of Multiple Metric Spaces

Each metric space directly addresses the limitations of single-space learning through complementary geometric perspectives:

- **Euclidean distance:** Measures absolute differences between feature vectors, effective for attacks with clear separability based on magnitude.
- **Cosine distance:** Captures angular similarity between samples, useful for detecting adversarial perturbations or slight variations in network traffic patterns.
- **Chebyshev distance:** Emphasizes the *largest feature deviation* across dimensions, helping to detect localized anomalies in specific network attributes.
- **Wasserstein distance**: Compares *entire statistical distributions*, enabling the detection of *subtle, aggregated deviations* in attack behaviors.

Each metric captures a different notion of similarity (geometry, direction, worst-case deviation, distributional shift). MSPL normalizes these distances and fuses them with learned/assigned weights to leverage their complementary biases within each episode. The integration of these spaces ensures that MSPL can effectively adapt to different *attack distributions and intrusion scenarios*, leading to improved generalization.

B. The Need for a Proper Metric Space Fusion

While individual metric spaces offer distinct geometric perspectives, their isolated application restricts the topological expressiveness necessary for comprehensive attack characterization. Integrating these metrics requires addressing fundamental challenges in distance synthesis. First, the numerical scales across metrics exhibit significant heterogeneity—Euclidean distances typically scale with $\sqrt{d}$ for d-dimensional data, while Cosine distances are bounded within $[0, 1]$, and Wasserstein distances reflect distributional properties with domain-specific magnitudes. These discrepancies create disproportionate gradient contributions during optimization, potentially destabilizing the learning process. Second, direct linear combination of distances can compromise the mathematical properties that ensure consistent prototype generation, particularly the triangle inequality essential for well-formed decision boundaries. Specifically, for any pair of samples x_i and x_j, the fusion process must preserve the relationship:

$$D(x_i, x_j) \leq D(x_i, x_k) + D(x_k, x_j) \quad \forall x_k \in \mathcal{X}, \qquad (1)$$

where x_i, x_j, and x_k represent individual data samples from the input space $\mathcal{X}$. The function $D(x_a, x_b)$ denotes the fused distance between two samples x_a and x_b, calculated as a weighted sum of normalized distances across multiple metric spaces. Each individual distance is first normalized using z-score normalization (with clipping), and then combined using non-negative weights w_m such that $\sum w_m = 1$, ensuring that the resulting fused distance maintains essential metric properties like the triangle inequality.

Third, attack patterns demonstrate contextual metric relevance. Some intrusions primarily manifest through direction shifts (favoring Cosine), while others show magnitude anomalies (favoring Euclidean) or localized feature deviations (favoring Chebyshev). Although a dynamic context-dependent weighting method might offer flexibility in highlighting the most informative metric for each attack type, our approach opts for a predetermined weighting strategy. In practice, we assign fixed, even-split weights—for instance, configurations such as (1, 0, 0, 0), (1/2, 1/2, 0, 0), or (0, 1/3, 1/3, 1/3)—to ensure a consistent contribution from each metric space according to the experimental setup.

C. Stability and Generalization in Few-Shot Learning

Network intrusion detection systems face heightened stability challenges in few-shot learning settings compared to conventional applications in computer vision or natural language processing. The fundamental issue arises from statistical irregularity; attack distributions exhibit multimodal characteristics with long-tailed distributions across feature dimensions. Consequently, when forming prototypes from extremely limited samples (often $K < 5$ for rare attack types), the resulting representations demonstrate high variance, particularly for zero-day attacks where emerging patterns may manifest only in specific feature subspaces [2].

This prototype variance problem manifests itself in three critical ways specific to network security:

- **Prototype instability**: Minor variations in the support set composition produce drastically different prototype vectors, particularly for statistically heterogeneous attacks like DDoS or port scanning.
- **Catastrophic forgetting**: Sequential learning of new attack patterns can disrupt previously established decision boundaries, a phenomenon exacerbated by the inherent volatility of episodic sampling.
- **Confidence calibration**: High variance in prototype computation leads to miscalibrated confidence scores, reducing the model's ability to differentiate between known attacks and genuinely novel intrusions.

While traditional regularization techniques (e.g., L_1/L_2 penalties, dropout) mitigate overfitting, they fail to address the systemic instability arising from support set variability. This requires specialized stabilization mechanisms, such as temporal smoothing through parameter averaging and balanced episodic construction, that specifically target the unique challenges of learning from minimal network attack samples.

IV. METHODOLOGY

The source code for our MSPL implementation is available at.[1]

A. Problem Formulation

Given a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, where $x_i \in \mathcal{X} \subseteq \mathbb{R}^d$ and $y_i \in \{1, \ldots, C\}$ is a *single-label (multi-class)* attack category. For notational convenience we also use its one-hot encoding $\mathbf{y}_i \in \{0, 1\}^C$ with $\sum_{k=1}^{C} y_{i,k} = 1$ (mutually exclusive classes). Our objective is to learn a robust embedding function $f_\theta : \mathcal{X} \to \mathbb{R}^m$ that maps traffic inputs into a representation space where multiple distance metrics collectively enhance intrusion pattern detection.

B. FSL for Attack Detection

In operational network environments, novel attack patterns frequently emerge with extremely limited samples, particularly in zero-day scenarios where attack signatures are initially unknown. Our framework addresses this FSL challenge through a C-way K-shot formulation, where C represents distinct attack classes, and K denotes the minimal number of labeled samples available per attack type. Formally, for each episodic task:

$$\mathcal{T} = \{(c, \mathcal{S}_c, \mathcal{Q}_c) | c \in \text{Attack Classes}\}$$

where $\mathcal{S}_c$ represents the support set containing K examples of attack class c, and $\mathcal{Q}_c$ is the query set for evaluation. The stratified sampling ensures the representation of rare attack patterns:

$$\mathbb{P}(|\mathcal{S}_c| \geq K_{min}) = 1, \forall c \in \text{Attack Classes}$$

This formulation is particularly crucial for:

- Zero-day attack detection with limited samples, enabling rapid response to emerging threats
- Attack variant identification despite adversarial modifications designed to evade detection
- Balanced learning across attack types regardless of their natural frequency distribution, mitigating bias toward common attack patterns at the expense of rare but potentially severe intrusions

C. Multi-Space Prototypical Framework

Our framework extends traditional prototypical networks through three key innovations:

1) Simultaneous operation across complementary metric spaces
2) Constrained metric space integration
3) Polyak-averaged prototype generation

[1] https://anonymous.4open.science/r/Dos-Project-325F

The framework operates across four carefully selected metric spaces $\mathcal{M} = \{E, C, Ch, W\}$, each capturing distinct topological properties:

1) Distance Metrics: For an embedding $f_\theta(x)$ and prototype c_k, we define:

1) Euclidean distance (d_E) to capture global geometric relationships: $d_E(x, c_k) = \|f_\theta(x) - c_k\|_2$.
2) Cosine distance (d_C) to measure directional similarities: $d_C(x, c_k) = 1 - \frac{f_\theta(x) \cdot c_k}{\|f_\theta(x)\| \|c_k\|}$.
3) Chebyshev distance (d_{Ch}) to identify maximum deviations: $d_{Ch}(x, c_k) = \max_i |f_\theta(x)_i - c_{k_i}|$.
4) Wasserstein distance (d_W) to capture distribution-level differences: $d_W(x, c_k) = \mathbb{E}_{U \sim \mathcal{U}(0,1)}[|F_x^{-1}(U) - F_{c_k}^{-1}(U)|]$.

2) Metric Space Integration: Each distance metric undergoes z-score normalization with clipping to ensure comparable scales:

$$\hat{d}_m(x) = \text{clip}\left(\frac{d_m(x) - \mu_m}{\max(\sigma_m, \epsilon)}, -\gamma, \gamma\right),$$

where ϵ prevents division by zero and γ controls the clipping range. Without normalization, the raw metric with the largest magnitude (typically Wasserstein, which scales $\mathcal{O}(10^2 - 10^3)$ compared to Cosine's [0,2] range) would dominate gradient contributions during backpropagation, effectively collapsing the multi-space fusion to single-metric behavior.

The integration follows a constrained weighting scheme:

$$D(x, c_k) = \sum_{m \in \mathcal{M}} w_m \cdot \hat{d}_m(x, c_k)$$

subject to:

$$\sum_{m \in \mathcal{M}} w_m = 1, \quad w_m \geq 0 \quad \forall m \in \mathcal{M}.$$

D. Model Stabilization Through Polyak Averaging

We tested Polyak averaging at the model level to improve stability and convergence in our few-sample setting. Rather than averaging prototypes directly, we maintain an exponential moving average (EMA) of the entire model parameters θ, which implicitly stabilizes the prototype generation process through a more robust embedding function f_θ.

For model parameters at training iteration t, the EMA parameters are updated as:

$$\theta_{\text{EMA}}^{(t)} = \beta \theta_{\text{EMA}}^{(t-1)} + (1 - \beta)\theta^{(t)},$$

where β controls the temporal decay of previous parameter values. This averaging provides three key benefits in our few-shot context: (1) stabilization of the embedding function f_θ across episodes, reducing variance in prototype computation, (2) implicit ensembling of models along the optimization trajectory, and (3) smoothing of the optimization landscape to avoid sharp minima that often lead to poor generalization.

The class prototypes c_k are then computed using the EMA model during inference as $c_k = \frac{1}{|\mathcal{S}_k|} \sum_{(x_i, y_i) \in \mathcal{S}_k} f_{\theta_{\text{EMA}}}(x_i)$.

E. Episodic Learning Framework

Our framework implements episodic training through stratified sampling to handle class imbalance in network attacks. The episode construction process begins with initial dataset sampling to ensure class representation:

$$(\mathcal{X}_{train}, \mathcal{Y}_{train}) = \text{stratified\_sample}(\mathcal{D}, n_{samples}).$$

Episodes $\mathcal{E} = \{(\mathcal{S}_i, \mathcal{Q}_i)\}_{i=1}^N$ are then constructed, where each episode contains support and query sets of fixed sizes:

$$|\mathcal{S}_k| = N_s \text{and} \quad |\mathcal{Q}_k| = N_q, \forall k \in \{1, \ldots, C\}.$$

To address the inherent class imbalance in attack detection, we employ an adaptive sampling mechanism. For each class k with insufficient samples, we utilize controlled repetition sampling:

$$\mathcal{S}_k = \text{sample}(\mathcal{D}_k \times \lceil (N_s + N_q)/|\mathcal{D}_k| \rceil, N_s + N_q).$$

If a class has at least $N_s + N_q$ distinct samples, we sample support/query without replacement and ensure $S_k \cap Q_k = \emptyset$. For low-count classes, we use controlled repetition and avoid overlap whenever possible; overlap may occur only when unavoidable.

For a query sample x, we use the negative fused distances as logits $\ell_k(x) = -D(x, c_k), \quad k \in \{1, \ldots, C\}$. The class posterior is computed with a softmax as $p_\theta(y = k \mid x) = \frac{\exp(\ell_k(x))}{\sum_{j=1}^C \exp(\ell_j(x))}$. We then optimize the multi-class cross-entropy over the query set Q:

$$\begin{aligned} \mathcal{L} &= -\sum_{(x,y) \in Q} \log p_\theta(y \mid x) \\ &= -\sum_{(x,\mathbf{y}) \in Q} \sum_{k=1}^C y_k \log p_\theta(y = k \mid x), \end{aligned} \tag{2}$$

At inference, MSPL predicts $\hat{y}(x) = \arg\max_k p_\theta(y = k \mid x)$ (equivalently, $\arg\min_k D(x, c_k)$); thus, *no thresholding is used* in the multi-class setting.

F. Training Procedure

The training process follows Algorithm 1, which implements our MSPL framework. The algorithm alternates between episode-based training and validation phases, maintaining model stability through gradient clipping and early stopping mechanisms.

For each training iteration, the process:

1) Computes embeddings for support and query sets through f_θ.
2) Generates class prototypes from support set embeddings.
3) Calculates and normalizes distances across all metric spaces.
4) Updates model parameters through gradient descent with clipping.

When Polyak averaging is enabled, θ_{EMA} is updated after each training parameter update, and only θ_{EMA} (not θ) is used for validation/model selection and inference.

V. Theoretical Guarantees and Model Properties

Our framework operates on a collection of metric spaces $\mathcal{M}$, where each metric $m \in \mathcal{M}$ satisfies the fundamental properties of non-negativity ($d_m(x, y) \geq 0$), symmetry ($d_m(x, y) = d_m(y, x)$), and triangle inequality ($d_m(x, z) \leq d_m(x, y) + d_m(y, z)$). To ensure comparable scales across

Algorithm 1 MSPL Training and Inference via Episodic Meta-Learning, Combining Normalized Multi-Metric Distances, Weighted Fusion, and Polyak (EMA) Averaging.

Input: Input: Dataset $\mathcal{D}$, episodes N_e, ways N_{way}, support size N_s, query size N_q, metric weights $\{w_m\}_{m\in\mathcal{M}}$, Polyak decay β
Output: Best model parameters θ^*

```
Initialize model parameters θ randomly
(X_train, Y_train) ← sample(D)
if using Polyak averaging then
    Initialize θ_EMA ← θ
end if
E ← CreateEpisodes(X_train, Y_train, N_e, N_way, N_s, N_q)
▷ Per episode: sample N_way classes; draw N_s support and N_q query per class.
▷ If |I_k| ≥ N_s+N_q: sample w/o replacement so S_k∩Q_k = ∅; else use controlled repetition (avoid overlap if possible).
for epoch = 1 to E do
    for each (S, Q) ∈ E do
        Z_s ← f_θ(S)
        Z_q ← f_θ(Q)
        {c_k} ← ComputePrototypes(Z_s)
        for each metric m ∈ M do
            d_m ← ComputeDistance(Z_q, {c_k}, m)
            d̂_m ← NormalizeDistance(d_m)
        end for
        D ← Σ_{m∈M} w_m · d̂_m
        L ← ComputeLoss(D, Q)
        Clip gradients and update θ using ∇_θ L
        if using Polyak averaging then
            θ_EMA ← βθ_EMA + (1 − β)θ
        end if
    end for
    Validate model and save if improved
end for
return θ*
```

different metrics, each distance undergoes z-score normalization with clipping, transforming raw distances into normalized metrics $\hat{d}_m$. The weighted combination of these normalized metrics, defined as $D(x,y) = \sum_{m\in\mathcal{M}} w_m \hat{d}_m(x,y)$ under the constraint $\sum_{m\in\mathcal{M}} w_m = 1, w_m \geq 0$, preserves metric properties while enabling flexible contributions from each space. This preservation is demonstrated through the triangle inequality:

$$\begin{aligned} D(x,z) &= \sum_{m\in\mathcal{M}} w_m \hat{d}_m(x,z) \\ &\leq \sum_{m\in\mathcal{M}} w_m(\hat{d}_m(x,y) + \hat{d}_m(y,z)) \\ &= D(x,y) + D(y,z) \end{aligned}$$

A. Convergence and Stability

The convergence properties of our framework are established under three key conditions: Lipschitz continuity of each normalized metric $\hat{d}_m$ with constant L_m, β-smoothness of the combined loss, and bounded gradients $\|\nabla_\theta \mathcal{L}\| \leq G$. Given these conditions and our episodic training procedure with Polyak averaging, the expected error convergence is bounded by the following:

$$\begin{aligned} &\mathbb{E}[\|\theta^{(T)} - \theta^*\|^2] \\ &\quad \leq \frac{1}{T}\left(\frac{\|\theta^{(0)} - \theta^*\|^2}{\eta} + \frac{\eta G^2}{2}\right) \cdot \left(\sum_{m\in\mathcal{M}} w_m L_m\right) \end{aligned}$$

This bound accounts for the contribution of each metric space through their respective weights and Lipschitz constants.

B. Space Complementarity

The effectiveness of our multi-space approach derives from the fundamental properties of metric space interactions and their collective contribution to learning discriminative representations.

1) Metric Space Interaction Properties: For any pair of metrics $m_1, m_2 \in \mathcal{M}$, their interaction is characterized by:

Property 1 (Metric Complementarity): For any two metrics $m_1, m_2 \in \mathcal{M}$, there exists a subset of points $\mathcal{X}_{m_1,m_2} \subset \mathcal{X}$ where:

$$|\hat{d}_{m_1}(x,y) - \hat{d}_{m_2}(x,y)| \geq \epsilon_{m_1,m_2}$$

for some $\epsilon_{m_1,m_2} > 0$, ensuring each metric contributes unique discriminative information.

This means that there are pairs of samples for which different metrics disagree by a nontrivial margin, so each metric can highlight distinctions that another metric may miss.

Property 2 (Multi-Metric Coverage): For any embedding $f_\theta(x)$, the combined metric space ensures comprehensive coverage:

$$\max_{m\in\mathcal{M}} \hat{d}_m(x,y) \geq \lambda \cdot d_{true}(x,y),$$

where d_{true} represents the true underlying distance and $\lambda > 0$.

In practice, at least one metric in the set provides a sufficiently strong separation that scales with the underlying notion of dissimilarity, preventing uniformly weak distances across all metrics.

2) Extensibility Theorem: The framework maintains its theoretical guarantees when extended with additional metrics that satisfy:

$$\forall m_{new} \in \mathcal{M}_{ext} : \exists \alpha_m > 0 \text{s.t.} \quad d_{m_{new}}(x,y) \geq \alpha_m \|x - y\|$$

In the context of attack detection, metric complementarity provides:

1) **Attack Pattern Coverage:** For attack class c and normal traffic n:

$$\exists m \in \mathcal{M} : \mathbb{P}(\hat{d}_m(x, c_c) < \hat{d}_m(x, c_n) | y = c) \geq 1 - \delta_m$$

2) **Multi-Faceted Attack Characterization:**

$$\text{characterization}(x) = [\hat{d}_E(x), \hat{d}_{Ch}(x), \hat{d}_C(x), \hat{d}_W(x)]$$

capturing different aspects of attack patterns.

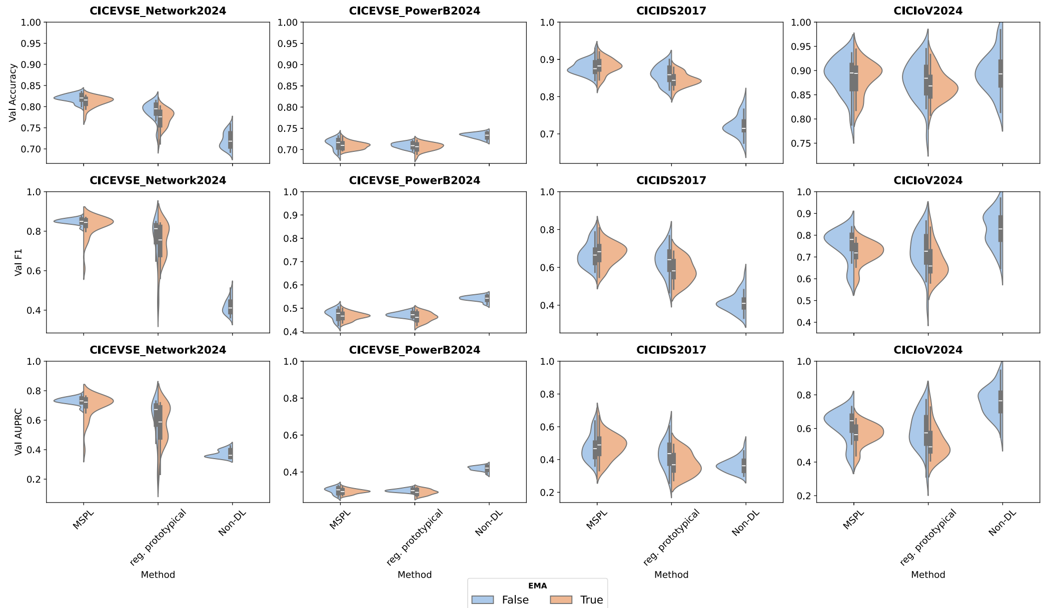

Fig. 1. Comparison of MSPL framework variants using different metric combinations (euclidean, chebyshev, cosine, and wasserstein) across three benchmark intrusion detection datasets. The results highlight the impact of metric fusion on detection accuracy, F1-score, and AUPRC, demonstrating the advantages of multi-space integration over single-metric baselines.

C. Multi-Space Optimization Properties

The optimization process in our framework maintains several key properties that ensure effective learning:

1) **Distance Normalization Stability:** For any metric $m \in \mathcal{M}$, the normalized distance satisfies:

$$\|\hat{d}_m(x, y)\|_\infty \leq \gamma,$$

where γ is the clipping threshold, ensuring bounded gradients during optimization.

2) **Prototype Convergence:** Given the bounded embeddings from f_θ and z-score normalized distances, the prototype convergence in expectation follows:

$$\mathbb{E}[\|c_k^{(t)} - c_k^*\|_2] \leq \frac{1}{\sqrt{|\mathcal{S}_k|}} \cdot \sigma_k,$$

where σ_k represents the intra-class variance and $|\mathcal{S}_k|$ is the support set size for class k.

D. Computational Complexity

The computational complexity of our framework for each episode is $\mathcal{O}(N_s N_q |\mathcal{M}|)$. This includes:

- Support set embedding computation:$\mathcal{O}(N_s)$
- Query set distance calculations:$\mathcal{O}(N_q)$
- Multi-space metric computations:$\mathcal{O}(|\mathcal{M}|)$

The memory complexity is bounded by $\mathcal{O}(N_s + N_q + |\mathcal{M}|C)$, where C is the number of classes.

VI. EVALUATION

This section comprehensively evaluates the proposed MSPL framework, demonstrating its efficacy in few-shot network intrusion detection.

The experiments employed three benchmark datasets with distinct contexts to assess performance under varied traffic scenarios: CICEVSE2024 [16] (EV charging-station network traffic/attacks), CICIDS2017 [17] (traditional enterprise network traffic with common attacks), and CICIoV2024 [18] (in-vehicle IoV CAN-bus traffic/attacks). To rigorously evaluate the FSL capabilities, we restricted the training to only 200 instances. This constrained setting mirrors real-world scenarios where labeled attack data is scarce. The evaluation emphasizes three key metrics, balanced accuracy, validation F1-score, and area under the precision-recall curve (AUPRC), providing insights into the framework's ability to generalize, detect rare attack types, and maintain high classification performance across imbalanced datasets. For AUPRC in the multi-class setting, we compute one-vs-rest precision-recall per class using the softmax score $p_\theta(y = k \mid x)$ and report the macro-average across classes.

A. Performance Analysis

1) Cicevse Datasets: On the CICEVSE Network2024 dataset, the baseline approach achieved a balanced accuracy of 0.7210, with a validation F1-score of 0.4194 and an AUPRC of 0.3719 (Table I, Baseline results, and Fig. 1). The MSPL framework significantly improved these metrics, increasing the balanced accuracy to 0.8200, raising the validation F1-score to 0.8502, and elevating the AUPRC to 0.7324 (Table I, MSPL results). Integrating tri-metric spaces, Euclidean, Chebyshev, and Cosine, provided complementary perspectives on intrusion patterns, contributing to substantial performance gains.

Each experiment was conducted over 40 random seeds. Results are reported as the mean $\pm$ 95% confidence interval (CI). The CIs are based on the Student's t-distribution with 39 degrees of freedom ($N = 40$), that is, $\bar{x} \pm t_{0.975,39} \cdot \frac{s}{\sqrt{40}}$, where $\bar{x}$ and s denote the sample mean and the sample standard deviation, respectively.

2) CICIDS2017 Dataset: For the CICIDS2017 dataset, the baseline (Regular Prototypical without Polyak) achieved a balanced accuracy of 0.8585, a validation F1-score of 0.6327,

TABLE I

FSL Attack Detection Performance Comparison Under FSL (200 Samples, 40 seeds). Best Results Per Dataset Are bolded. Shaded Rows Indicate Our MSPL Framework

Dataset	Method	Metric Space*	Polyak	Bal. Acc	F1-Score	AUPRC
CICEVSE Network	Non-DL Baseline	—	—	0.7210 ± 0.0061	0.4194 ± 0.0122	0.3719 ± 0.0082
	Reg. Prototypical	E	×	0.7905 ± 0.0075	0.7677 ± 0.0392	0.6145 ± 0.0494
	Reg. Prototypical	E	✓	0.7713 ± 0.0080	0.7374 ± 0.0365	0.5705 ± 0.0481
	MSPL (Ours)	E + Ch + C	×	**0.8200 ± 0.0034**	**0.8502 ± 0.0052**	**0.7324 ± 0.0084**
	MSPL (Ours)	E + Ch + C	✓	0.8112 ± 0.0048	0.8263 ± 0.0202	0.6964 ± 0.0294
CICEVSE PowerB	Non-DL Baseline	—	—	0.7329 ± 0.0020	0.5420 ± 0.0036	0.4187 ± 0.0041
	Reg. Prototypical	C	×	0.7097 ± 0.0025	0.4737 ± 0.0040	0.2997 ± 0.0032
	Reg. Prototypical	E	✓	0.7060 ± 0.0035	0.4616 ± 0.0071	0.2903 ± 0.0055
	MSPL (Ours)	Ch + C	×	**0.7388 ± 0.0048**	**0.5521 ± 0.0086**	**0.4288 ± 0.0048**
	MSPL (Ours)	Ch + C	✓	0.7298 ± 0.0032	0.5460 ± 0.0061	0.4268 ± 0.0048
CICIDS2017	Non-DL Baseline	—	—	0.7229 ± 0.0091	0.4147 ± 0.0162	0.3669 ± 0.0139
	Reg. Prototypical	E	×	0.8585 ± 0.0091	0.6327 ± 0.0278	0.4319 ± 0.0327
	Reg. Prototypical	E	✓	0.8453 ± 0.0066	0.5826 ± 0.0232	0.3729 ± 0.0255
	MSPL (Ours)	E + W + C	×	0.8786 ± 0.0070	0.6691 ± 0.0227	0.4749 ± 0.0294
	MSPL (Ours)	E + W + C	✓	**0.8806 ± 0.0078**	**0.6731 ± 0.0226**	**0.4799 ± 0.0286**
CICIoV2024	Non-DL Baseline	—	—	**0.8955 ± 0.0130**	**0.8274 ± 0.0253**	**0.7615 ± 0.0279**
	Reg. Prototypical	E	×	0.8804 ± 0.0150	0.7321 ± 0.0328	0.5881 ± 0.0409
	Reg. Prototypical	E	✓	0.8691 ± 0.0107	0.6805 ± 0.0283	0.5219 ± 0.0355
	MSPL (Ours)	Ch + C	×	0.8875 ± 0.0140	0.7540 ± 0.0267	0.6144 ± 0.0338
	MSPL (Ours)	E + C	✓	0.8848 ± 0.0124	0.7160 ± 0.0195	0.5626 ± 0.0237

* **Metric Space Legend:** E → Euclidean, C → Cosine, Ch → Chebyshev, W → Wasserstein. Weights are distributed equally among active metrics.

and an AUPRC of 0.4319 (Table I). The MSPL framework improved these results by attaining a balanced accuracy of 0.8806, F1-score of 0.6731, and AUPRC of 0.4799. The use of a tri-metric approach, combining Euclidean, Wasserstein, and Cosine metrics, enabled better generalization across attack types, resulting in robust and consistent improvements. This improvement indicates higher sensitivity under CICIDS2017's long-tailed class distribution, i.e., fewer missed detections for low-support attack categories in the few-shot regime.

3) CICIoV2024 Dataset: The CICIoV2024 dataset exhibited strong baseline results with a balanced accuracy of 0.8804, F1-score of 0.7321, and AUPRC of 0.5881 (Regular Prototypical without Polyak). The MSPL approach delivered further improvements, with balanced accuracy of 0.8875, F1-score of 0.7540, and AUPRC of 0.6144. The use of a bi-metric space combining Chebyshev and Cosine metrics demonstrated adaptability, particularly in addressing class imbalance and subtle attack signatures.

4) Non-DL Baseline: Besides deep-learning-based approaches, we experimented with traditional machine learning models. Table I shows that for *CICEVSE-Network2024*, performance was relatively low (Balanced Accuracy: 0.7210, F1: 0.4194, AUPRC: 0.3719), and similarly for *CICEVSE-PowerB2024* (Balanced Accuracy: 0.7329, F1: 0.5420, AUPRC: 0.4187) and *CICIDS2017* (Balanced Accuracy: 0.7229, F1: 0.4147, AUPRC: 0.3669). In contrast, performance was highest in *CICIoV2024* (Balanced Accuracy: 0.8955, F1: 0.8274, AUPRC: 0.7615). These results indicate that traditional approaches struggled with more complex or imbalanced datasets, while performing reasonably well on simpler ones. This gap is likely due to the higher heterogeneity and imbalance of EV-charging traffic in CICEVSE under few-shot supervision, whereas CICIoV2024 traces are more structured/regular, yielding more separable attack patterns for feature-based classifiers.

B. Polyak Averaging Effectiveness

The effectiveness of Polyak averaging was evaluated across the datasets, revealing dataset-specific impacts. For the CICEVSE datasets, Polyak averaging introduced minimal variations, with performance metrics remaining stable (Table I, Polyak rows). In the case of CICIoV2024, it contributed to marginal stabilization in validation metrics, reflecting modest gains in reliability. The CICEVSE dataset exhibited the most significant benefits from Polyak averaging, with noticeable stabilization and smoothing effects (Table I, Polyak rows), highlighting its utility in high-performing environments.

C. Metric Space Contribution

The transition from a single-metric Euclidean baseline to a multi-space approach underlines the importance of metric complementarity. On the CICEVSE Network2024 dataset, the tri-metric approach enhanced pattern recognition by leveraging distinct geometric and directional properties (Table I, Metric Space column). For CICIDS2017, the tri-metric configuration (Euclidean, Wasserstein, Cosine) yielded consistent gains across the reported metrics (Table I), including AUPRC (0.4799 vs. 0.4319) and balanced accuracy (0.8806 vs. 0.8585). For CICIoV2024, the bi-metric configuration (Chebyshev, Cosine) achieved the best MSPL results on that dataset, improving AUPRC from 0.5881 to 0.6144 and balanced accuracy from 0.8804 to 0.8875, indicating that the optimal metric combination can be dataset dependent.

D. Key Observations and Discussions

1) Generalizability: The MSPL framework demonstrated robust generalization capabilities in all datasets: CICEVSE

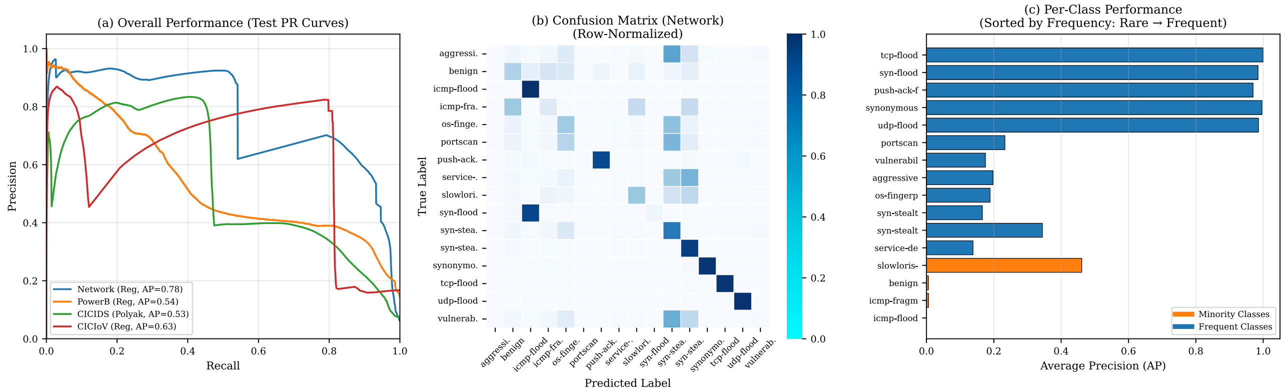

Fig. 2. Imbalanced few-shot conditions. (**left**) mean precision-recall curve over four datasets show stable performance, with relatively high AUPRC on CICEVSE network (0.78) and CICIoV (0.63). (**Center**) row-normalized confusion matrix on CICEVSE network data set shows significant diagonal, validating that it can still distinguish the different types of attacks, even with few training samples. (**Right**) Per-class AP, sorted by test set class frequency (rare → frequent).

Network2024, CICIDS2017, and CICIoV2024. For CICIDS2017, balanced accuracy improved from 0.8585 to 0.8806 and AUPRC from 0.4319 to 0.4799. For CICEVSE Network2024, balanced accuracy rose from 0.7210 to 0.8200 and AUPRC from 0.3719 to 0.7324. Even on the high-performing CICIoV2024 dataset, MSPL achieved further gains from 0.8804 to 0.8875 in balanced accuracy and 0.5881 to 0.6144 in AUPRC. These improvements validate the framework's ability to generalize across heterogeneous traffic profiles.

2) FSL Adaptation: The MSPL framework excelled in limited-sample environments, particularly on datasets with low baseline performance. In CICEVSE Network2024, AUPRC improved from 0.3719 to 0.7324, and validation F1-score increased from 0.4194 to 0.8502. These improvements highlight the power of episodic training and multi-metric fusion in learning from sparse data and enhancing adaptation to rare attacks.

3) Metric Space Complementarity: Integrating multiple distance metrics enhanced detection accuracy. In CICIDS2017, a tri-metric design (Euclidean, Wasserstein, Cosine) improved AUPRC (0.4799 vs. 0.4319). In CICIoV2024, a bi-metric approach (Chebyshev, Cosine) raised AUPRC from 0.5881 to 0.6144. In CICEVSE Network2024, a tri-metric space (Euclidean, Chebyshev, Cosine) boosted all metrics. These results show that diverse metrics capture distinct intrusion behaviors, enabling MSPL to model threats more effectively.

4) Per-Dataset Interpretation: The strongest practical impact is observed on CICEVSE (EV-charging), where the large AUPRC/F1 gains suggest MSPL substantially improves the precision-recall tradeoff under severe class imbalance, enabling more reliable detection of rare attacks in the few-shot regime. On CICIDS2017 (enterprise traffic), improvements are smaller but consistent, indicating MSPL primarily enhances robustness across diverse attack types when baseline separability is already high. On CICIoV2024 (in-vehicle CAN bus), the baseline performance is strong and MSPL yields modest additional gains, implying better capture of subtle CAN-bus attack signatures without requiring additional labeled data.

5) Online SDN/NFV Integration: MSPL can run online by embedding each record x as $z = f_{\theta_{\mathrm{EMA}}}(x)$ and scoring fused prototype distances $D(z, c_k)$ to output a class/confidence, which an SDN/NFV controller can map to actions (drop/deny, throttling, rerouting/steering, or SFC updates). End-to-end delay can be expressed as $L_{\mathrm{dec}} = L_{\mathrm{feat}} + L_{\mathrm{infer}} + L_{\mathrm{ctrl}}$ (controller-dependent); closed-loop latency/throughput benchmarking with a specific controller is left for future work.

E. Class-Imbalance Resilience

Figure 2 provides more detailed performance breakdown under our strict few-shot settings. PR curves (left) show strong detection performance across all the datasets we have benchmarked, with AUPRC up to 0.78 on CICEVSE Network. Importantly, the class-wise evaluation on this dataset shows that MSPL can learn distinguishable signatures for minority classes (e.g. slowloris, AP 0.45) in the presence of high data imbalance. The heavy diagonal in the confusion matrix (center) indicates that most attack types are predicted with low error rates, while the lower benign class performance is more likely due to its extreme sparsity (<0.01%) on this attack-heavy dataset rather than model breakdown.

VII. Conclusion

We proposed Multi-Space Prototypical Learning (MSPL) for few-shot attack detection, targeting emerging and rare intrusions under limited labeled data. By integrating complementary metric spaces (Euclidean, Cosine, Chebyshev, and Wasserstein), MSPL captures diverse geometric and distributional cues, while Polyak-averaged prototype generation improves stability and episodic training promotes balanced adaptation across classes and datasets. Experiments show consistent gains in balanced accuracy and AUPRC over strong baselines, and our analyses confirm the benefits of metric complementarity and scalability across diverse intrusion scenarios. From a deployment standpoint, MSPL is lightweight and episodic, making it practical to integrate into SDN/NFV-based security controllers for near real-time detection and response. Future extensions will explore multimodal network data (e.g., flows, logs, and host telemetry) and real-time integration with SDN/NFV platforms for online adaptation and intrusion prevention.

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